

Homework 7

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Define $\mathfrak{g} \subseteq \mathfrak{gl}(3, \mathbb{R})$ as the Lie $\mathbb{R}$ -algebra of matrices of the form

$$\begin{pmatrix} A & v \\ 0 & t \end{pmatrix}, \quad A \in \mathfrak{gl}(2, \mathbb{R}), v \in \mathbb{R}^2, t \in \mathbb{R}, 0 \in \mathbb{R}^2.$$

(1) Let I be the 2×2 identity matrix. We claim that the solvable radical $\mathfrak{r}$ of $\mathfrak{g}$ is

$$\mathfrak{r} = \left\{ \begin{pmatrix} cI & v \\ 0 & t \end{pmatrix} \mid c, t \in \mathbb{R}; v \in \mathbb{R}^2 \right\}.$$

First we show this is an ideal:

$$\begin{aligned} \left[\begin{pmatrix} A & w \\ 0 & s \end{pmatrix}, \begin{pmatrix} cI & v \\ 0 & t \end{pmatrix} \right] &= \begin{pmatrix} cA & Av + tw \\ 0 & st \end{pmatrix} - \begin{pmatrix} cA & cw + sv \\ 0 & st \end{pmatrix} \\ &= \begin{pmatrix} 0 & (A - sI)v - (c - t)w \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

This is of the form

$$\begin{pmatrix} 0 & u \\ 0 & 0 \end{pmatrix}$$

which is an element of $\mathfrak{r}$. Thus $\mathfrak{r}$ is an ideal of $\mathfrak{g}$. Additionally, this is also an element of the strictly upper triangular matrices $\mathfrak{n}$, which are known to be solvable. Thus $[\mathfrak{r}, \mathfrak{r}] \subseteq \mathfrak{n}$ implies that $\mathfrak{r}$ is also solvable.

To show that $\mathfrak{r}$ is maximal, let $\mathfrak{R}$ be the maximal solvable ideal in $\mathfrak{g}$. Then $\mathfrak{r} \subseteq \mathfrak{R}$. Additionally, $\mathfrak{R}/\mathfrak{r}$ is an ideal of the quotient $\mathfrak{g}/\mathfrak{r}$. It suffices to show that $\mathfrak{R}/\mathfrak{r} \cong 0$ in the quotient. By part (3), the quotient is semisimple, so the statement follows. We will not use maximality of $\mathfrak{r}$ in the proofs to parts (2) or (3), so this is not circular reasoning.

(2) Let

$$X := \begin{pmatrix} A & v \\ 0 & t \end{pmatrix} \in \mathfrak{g}.$$

We can decompose X as

$$X = X_A + X_d = \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & v \\ 0 & t \end{pmatrix}.$$

In the quotient, the second factor vanishes, meaning any equivalence class in the quotient can be represented by an element of the form

$$X_A = \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \quad A \in \mathfrak{gl}_2(\mathbb{R}).$$

We can thus identify $\mathfrak{g}/\mathfrak{r}$ with a Lie subalgebra of $\mathfrak{gl}_2(\mathbb{R})$ (it is not immediately clear that we can choose an identification such that this is a Lie subalgebra). Henceforth, we abuse notation and

identify X_A and A . However, this representation is not unique, since we can pull out a scalar multiple of the identity. In particular, we can pull out a factor of the trace:

$$A = \left(A - \left(\frac{1}{2} \text{tr} A \right) I \right) + \left(\frac{1}{2} \text{tr} A \right) I.$$

Thus $A = A - \frac{1}{2}(\text{tr} A)I$ when passing to the quotient, yielding a traceless representation for the class $[A]$ of A in the quotient. Note also that the traceless representation of $[A]$ is *unique*. Thus we may identify $[A]$ with its traceless representation, yielding a map $\varphi : \mathfrak{g}/\mathfrak{r} \rightarrow \mathfrak{sl}_2(\mathbb{R})$. The map is evidently an isomorphism, and moreover is an isomorphism of Lie algebras since the Lie brackets are both the commutator, and φ clearly preserves multiplication.

(3) $\mathfrak{sl}_2(\mathbb{R})$ is semisimple.

(4) Define $\mathfrak{l} := \mathfrak{sl}_2(\mathbb{R})$. Then parts (1), (2), and (3) show

$$\mathfrak{l} \cong \mathfrak{g}/\mathfrak{r}.$$

As $\mathbb{R}$ -vector spaces, it then follows that $\mathfrak{g} \cong \mathfrak{l} \oplus \mathfrak{r}$. Consider the semidirect product $\mathfrak{l} \ltimes \mathfrak{r}$ with respect to the adjoint representation $X \mapsto [X, -]_{\mathfrak{g}}$. The bracket is then given by

$$[(A, X), (B, Y)] = ([A, B]_{\mathfrak{l}}, [X, Y]_{\mathfrak{r}} + [A, Y]_{\mathfrak{g}} - [B, X]_{\mathfrak{g}}).$$

The isomorphism of $\mathbb{R}$ -vector spaces $\mathfrak{g} \cong \mathfrak{l} \oplus \mathfrak{r}$ can be given by

$$(A, X) \mapsto \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} + X,$$

since $\mathfrak{sl}_2(\mathbb{R}) \cap \mathfrak{r} = 0$ in $\mathfrak{g}$. The Lie bracket can thus be rewritten as

$$\begin{aligned} & [A, B]_{\mathfrak{l}} + [X, Y]_{\mathfrak{r}} + [A, Y]_{\mathfrak{g}} - [B, X]_{\mathfrak{g}} \\ &= \left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \right]_{\mathfrak{g}} + [X, Y]_{\mathfrak{g}} + \left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, Y \right]_{\mathfrak{g}} - \left[\begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix}, X \right]_{\mathfrak{g}} \\ &= \left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \right]_{\mathfrak{g}} + [X, Y]_{\mathfrak{g}} + \left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, Y \right]_{\mathfrak{g}} + \left[X, \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \right]_{\mathfrak{g}} \\ &= \left[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} + X, \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} + Y \right]_{\mathfrak{g}}. \end{aligned}$$

Therefore the isomorphism $\mathfrak{g} \oplus \mathfrak{l} \oplus \mathfrak{r}$ promotes to an isomorphism of Lie algebras $\mathfrak{g} \cong \mathfrak{l} \ltimes \mathfrak{r}$.